Predicting Property Values: A Python Machine Learning Journey

Transforming real estate valuation through artificial intelligence and data science

Understanding the Problem

Why We Need AI-Powered Property Valuation

Manual Appraisals Are Inconsistent

Human appraisers can vary significantly in their valuations of the same property

Market Complexity

Bhopal's real estate market has multiple factors affecting prices across different localities

Our Solution

A Python-based predictor that uses locality data and property characteristics to estimate prices accurately

Location

Neighborhood, schools, transport

Size

Area, bedrooms, usable space

Amenities

Facilities, upgrades, parking

Market Conditions

Supply, demand, recent sales

Our AI-powered system integrates diverse data points to overcome the inherent inconsistencies and complexities of traditional property valuation methods, providing a more reliable and accurate estimate.

Step 1: Market Data - Bhopal 2026 Rates

Setting Up the Foundation: Locality-Based Pricing

The first step in our predictor is establishing baseline market rates for different Bhopal localities. These rates represent the average price per square foot in 2026.

Locality	Rate/sq.ft	Segment
Arera Colony	₹6,650	Posh segment
Bawadia Kalan	₹5,650	Premium high-growth
Kolar Road	₹4,030	Evolving residential
Ayodhya Bypass	₹4,330	Modern townships
Hoshangabad Road	₹3,920	Major connectivity hub

These rates form the `base_rate` in our prediction function. Different localities command different prices based on development, connectivity, and demand.

Step 2: The Prediction Function

How We Calculate Property Prices

The `predict_price()` function is the core of our system. It takes three inputs and returns an estimated price in Lakhs.

01

Input Parameters:

- `area`: Selected locality (e.g., 'Arera Colony')
- `sqft`: Total area in square feet
- `bhk`: Number of bedrooms/halls/kitchens

02

Base Rate Lookup:

- Gets the `base_rate` for the selected locality
- Default fallback: ₹3,500/sq.ft if locality not found

03

BHK Premium Calculation:

- Adds a 5% premium for every BHK above 2
- Formula: `bhk_premium = 1 + (max(0, bhk - 2) * 0.05)`
- Example: 2 BHK = 1.0x, 3 BHK = 1.05x, 4 BHK = 1.10x

04

Final Price Calculation:

- Multiplies: `base_rate × sqft × bhk_premium`
- Divides by 100,000 to convert to Lakhs
- Formula: `total_price = (base_rate * sqft * bhk_premium) / 100000`

Practical Example: Let's predict the price for a 3 BHK property in Arera Colony with 1000 sq.ft.
Base Rate for Arera Colony: ₹6,650/sq.ft.
BHK Premium for 3 BHK: `1 + (max(0, 3 - 2) * 0.05) = 1 + (1 * 0.05) = 1.05`
Total Price: `(6650 * 1000 * 1.05) / 100000 = 69.825` Lakhs.

Step 3: User Interface & Input Collection

Making It Interactive: Collecting User Input

Our predictor is designed to be user-friendly. It guides users through selecting a locality and entering property details.

01

Display Welcome Message

--- 🏛️ Bhopal Real Estate AI Predictor (2026) ---

Select a locality from the options below:

02

List Available Localities

- Dynamically displays all 5 localities with numbered options
- User sees: 1. Arera Colony, 2. Bawadia Kalan, etc.

03

Collect User Inputs

- User selects locality by entering a number (1-5)
- User enters total area in square feet
- User enters number of BHK

04

Process & Display Result

- System calculates the predicted price
- Shows formatted output: "✅ Predicted Price for a 3 BHK in Arera Colony: ₹69.83 Lakhs"

The system ensures a clear, guided process from input to final prediction, making property valuation accessible.

Complete Code Implementation

The Full Python Solution

```python
import pandas as pd
from sklearn.linear_model import LinearRegression

# 1. 2026 Bhopal Market Data
bhopal_rates = {
    'Arera Colony': 6650,
    'Bawadia Kalan': 5650,
    'Kolar Road': 4030,
    'Ayodhya Bypass': 4330,
    'Hoshangabad Road': 3920
}

# 2. Prediction Function
def predict_price(area, sqft, bhk):
    base_rate = bhopal_rates.get(area, 3500)
    bhk_premium = 1 + (max(0, bhk - 2) * 0.05)
    total_price = (base_rate * sqft * bhk_premium) / 100000
    return total_price

# 3. User Interface
print("--- Bhopal Real Estate AI Predictor (2026) ---")
colonies = list(bhopal_rates.keys())
for i, colony in enumerate(colonies, 1):
    print(f"{i}. {colony}")

choice = int(input("Enter the number of your choice: "))
selected_area = colonies[choice - 1]
sqft = int(input("Enter total area in Square Feet: "))
bhk = int(input("Enter number of BHK: "))

price = predict_price(selected_area, sqft, bhk)
print(f"Predicted Price: Rs {price:.2f} Lakhs")
```

Real-World Example: Step-by-Step Calculation

Let's Predict a Property Price Together

A buyer wants to know the price of a 3 BHK property in Bawadia Kalan with 1,200 square feet.

Show the calculation step-by-step:

01

Step 1: User Input

- Locality: Bawadia Kalan (choice = 2)
- Area: 1,200 sq.ft
- BHK: 3

02

Step 2: Get Base Rate

- `base_rate` = `bhopal_rates['Bawadia Kalan']` = ₹5,650/sq.ft

03

Step 3: Calculate BHK Premium

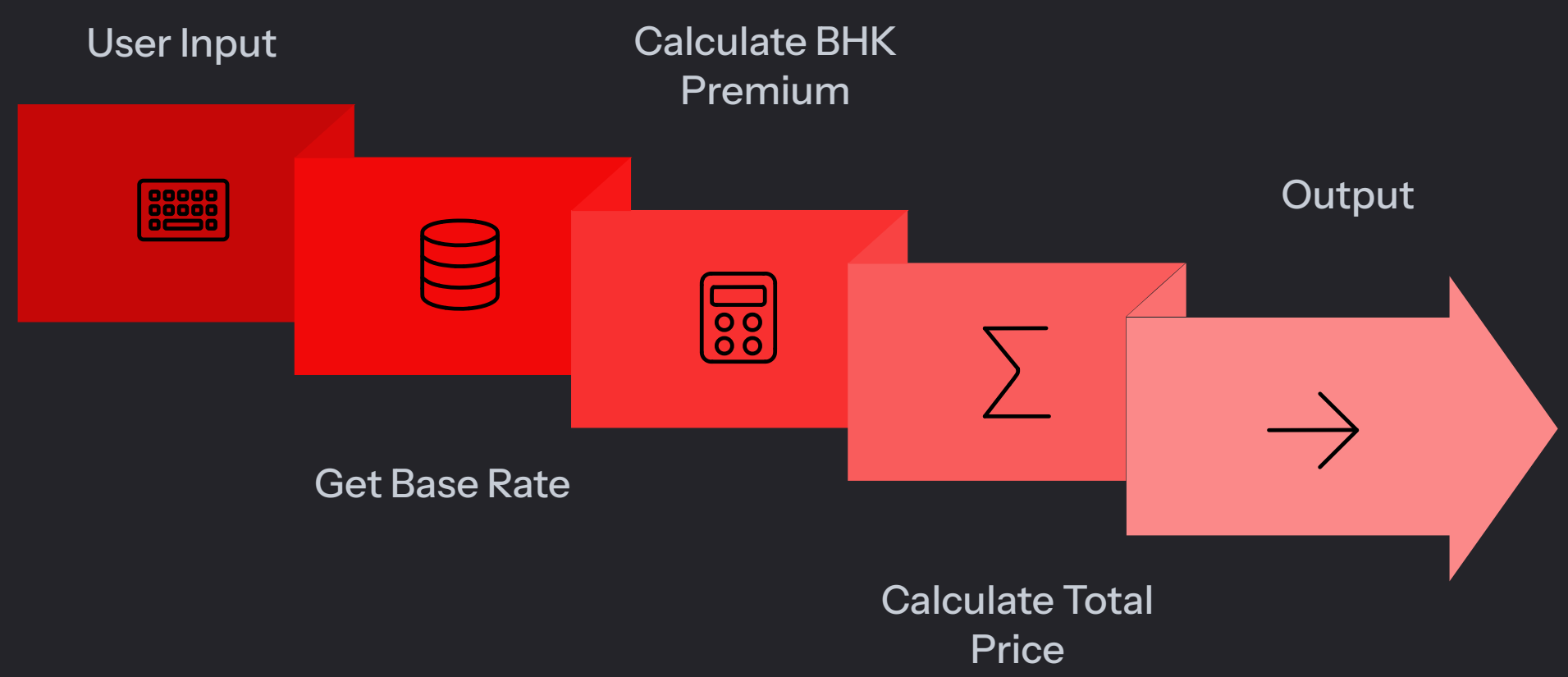
- `bhk_premium` = 1 + (max(0, 3 - 2) * 0.05)
- `bhk_premium` = 1 + (1 * 0.05) = 1.05

04

Step 4: Calculate Total Price

- `total_price` = (5650 * 1200 * 1.05) / 100000
- `total_price` = 7,119,000 / 100000
- `total_price` = 71.19 Lakhs

Output: "✅ Predicted Price for a 3 BHK in Bawadia Kalan: ₹71.19 Lakhs"



This visual summarizes the flow of data and calculation steps involved in predicting the property price, from initial user input to the final result.

Key Features & Advantages

Why This Approach Works

Data-Driven Accuracy

- Uses real 2026 Bhopal market rates
- Accounts for locality-specific pricing variations
- Incorporates property size and configuration

Intelligent Premium System

- BHK premium rewards larger properties
- 5% increase per additional bedroom above 2
- Reflects market demand for spacious homes

User-Friendly Design

- Simple numbered menu for locality selection
- Clear input prompts
- Formatted output with visual indicators (✅)

Scalability & Flexibility

- Easy to add new localities to bhopal_rates dictionary
- Can adjust premium percentages based on market trends
- Default fallback rate ensures robustness

Real-Time Predictions

- Instant calculations
- No need for external API calls
- Works offline with local data